

Exp no :9 Using Process Explorer to Identify Suspicious Processes

Date :

Description

- Process Explorer is a part of the Microsoft Sysinternals Suite. It is a powerful tool used to view detailed information about system processes. It helps investigators and administrators analyze active processes, detect suspicious behavior, monitor CPU and memory usage, and verify process authenticity using digital signatures.

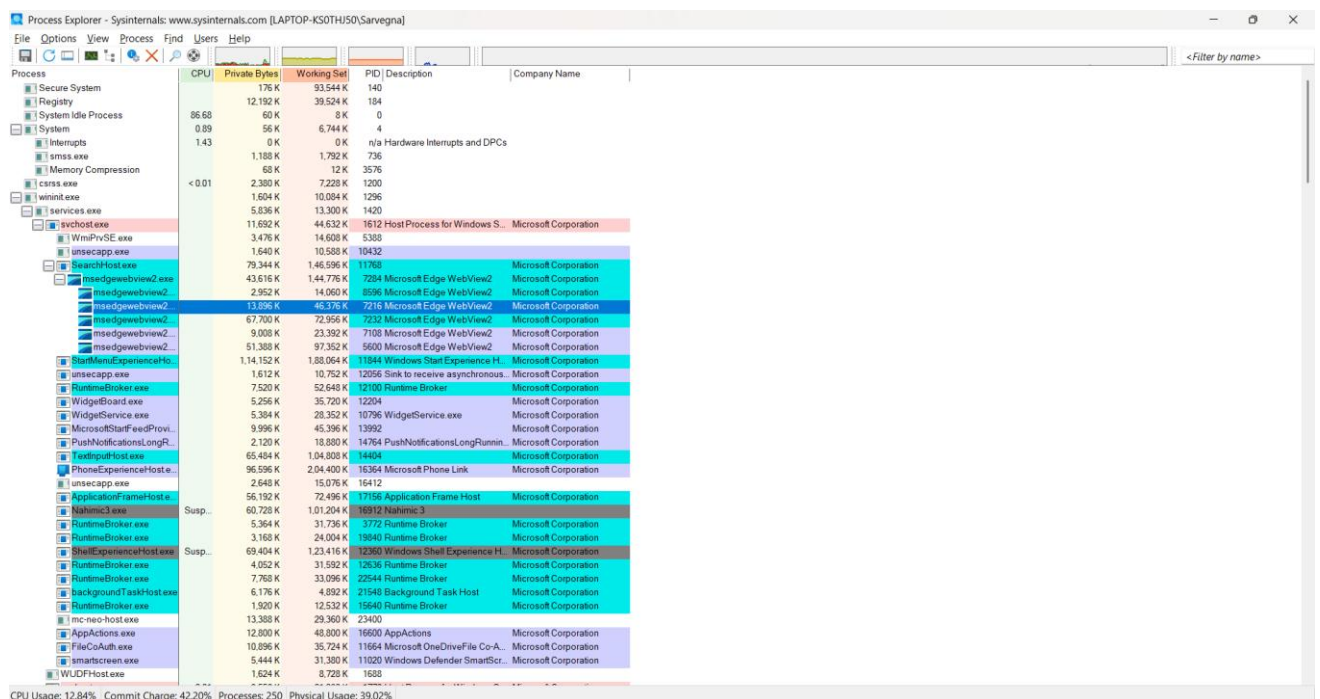
Step-by-Step Process

Step 1: Download and Setup Process Explorer

- Go to the official Microsoft Sysinternals website: <https://learn.microsoft.com/en-us/sysinternals/downloads/process-explorer>
- Click Download Process Explorer.
- Extract the downloaded ZIP file to a folder.
- Right-click procexp64.exe (for 64-bit) or procexp.exe (for 32-bit) → select Run as Administrator.

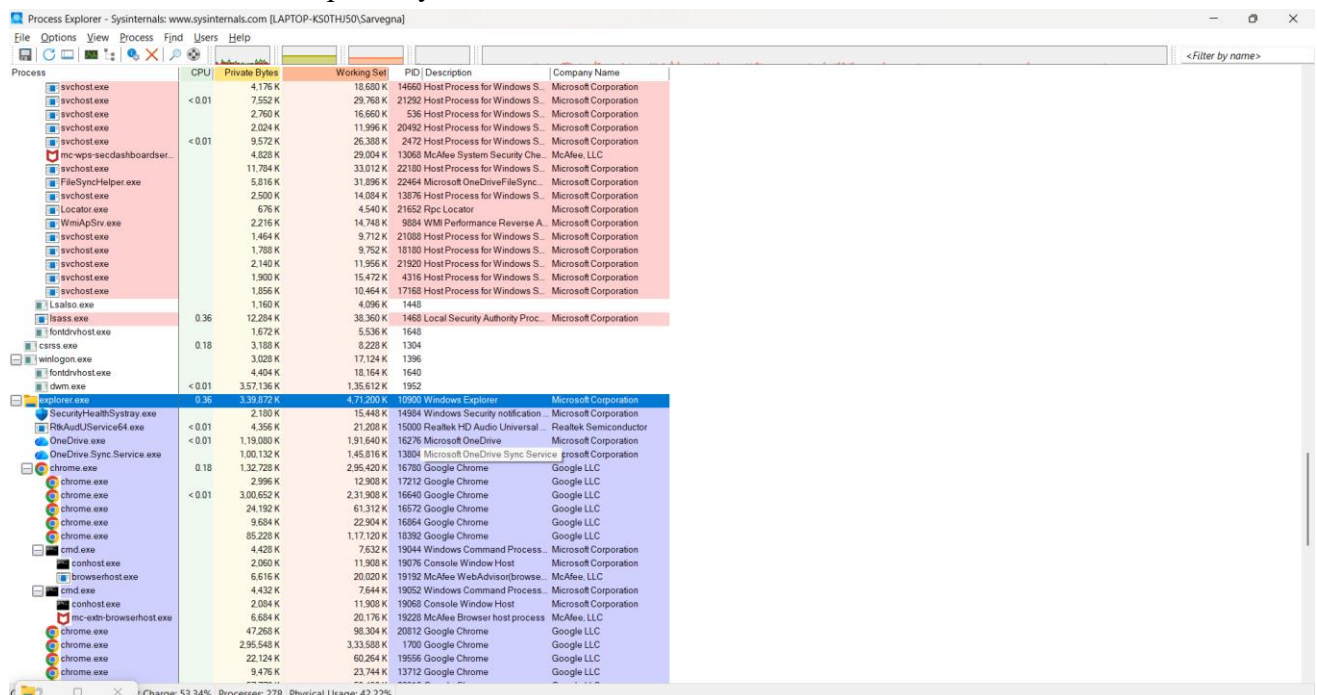
Step 2: Understand the Interface

- The main window displays all running processes in a hierarchical tree view.
- Each process shows details such as PID, CPU usage, memory usage, and company name.
- Color codes represent process states:
 - Green — Newly started processes
 - Red — Terminated processes
 - Light Blue — Processes running under the current user
 - Pink — Suspended processes



Step 3: Identify Suspicious Processes

1. Look for unfamiliar or oddly named processes (e.g., xkdjeo.exe, randomname123.exe).
2. Check the Company Name and Description:
 - Legitimate software usually shows known publishers like Microsoft, Intel, or Adobe.
3. Right-click the process → Properties → go to the Image tab.
4. Verify the Path of the executable file:
 - Safe: C:\Windows\System32\ ○ Suspicious: C:\Users\<User>\AppData\Temp\ or Downloads\
5. Check for Digital Signature:
 - Valid signature = trusted developer ○ No signature or invalid = possibly malicious



Step 4: Analyze Process Behavior

- Observe CPU, Memory, and I/O usage columns.
- If a small or unknown process consumes excessive CPU or memory, it may be malicious.
- Right-click the process → Properties → go to the TCP/IP tab.
- Check if it communicates with unknown external IP addresses.
- Examine Handles and DLLs tabs for suspicious loaded files or libraries.

explorer.exe:9748 Properties

name: Performance Performance Graph Disk and Network GPU Graph Threads TCP/IP Security Environment Job Strings

144			
TID	CPU	Cycles Delta	Suspend Count
10724	< 0.01	26,63,233	
10660			shcore.dllSHCreateThrea...
12148			windows.immersiveshell.s...
12372			ntdll.dllTpCallbackMayRu...
8604			Windows.UI.Xaml.dllGetE...
11768			shcore.dllSHCreateThrea...
21456			ntdll.dllTpCallbackMayRu...
1804			shcore.dllSHSetValueW+
9012			CoreMessaging.dllCoreUI...
12988			wlanapi.dllWlanPrivateCa...
13104			wlanapi.dllWlanPrivateCa...
2472			shcore.dllSHCreateThrea...
13052			wlanapi.dllWlanPrivateCa...
21940			ntdll.dllTpCallbackMayRu...
10888			ucrtbase.dllwschr+0x120
3204			shcore.dllSHSetValueW+
16816			shcore.dllSHSetValueW+
7044			shcore.dllSHSetValueW+
12324			shcore.dllSHCreateThrea...
12708			shcore.dllSHCreateThrea...
12308			ntdll.dllTpCallbackMayRu...
12312			ntdll.dllTpCallbackMayRu...
12300			ntdll.dllTpCallbackMayRu...

9976			
19:49:20	21-09-2026		
Wait:WtUserRequest		8	
0:00:07.812		12	
0:00:12.812		Normal	
95,533		5	
52,60,80,17,360		2	

Step 5: Verify Process Legitimacy

1. Search the process name on Google.
- Example: svchost.exe vs svhost.exe (one letter missing — suspicious).
2. Visit <https://www.virustotal.com> o Upload the process file or search its name to verify if it's reported as malware.
 3. Cross-check with ProcessLibrary.com or official vendor websites for authenticity.

Analysing (57.2s) ...

607fc0cc3787f53af7949bb93ccefbae0157aa218b2d43fe36cb1bc95eb8d0d

Size
47.06 MB

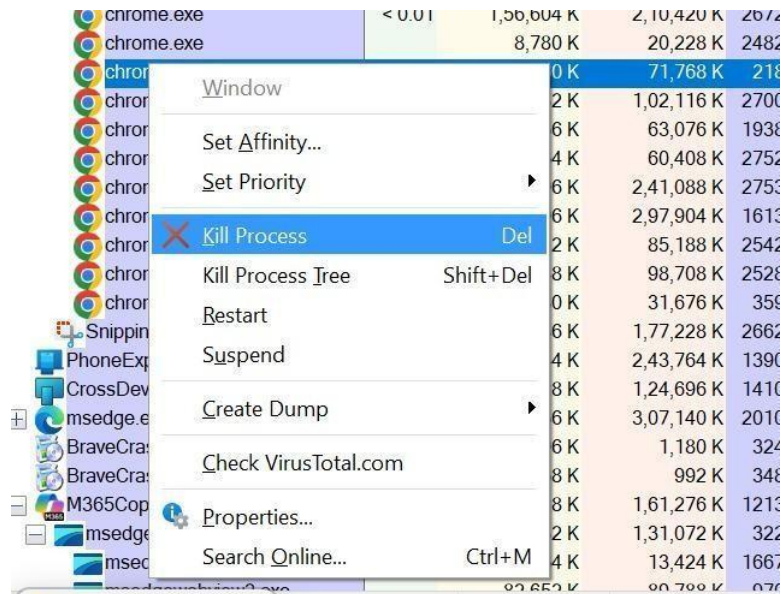
DETECTION

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis			
Acronis (Static ML)	Undetected	AhnLab-V3	Undetected
AliCloud	Undetected	ALYac	Undetected
Antiy-AVL	Undetected	Arcabit	Undetected
Arctic Wolf	Undetected	Avast	Undetected
Avira (no cloud)	Undetected	BitDefender	Undetected

Step 6: Take Appropriate Action

1. If the process is confirmed malicious:
 - o Right-click the process → Kill Process to stop it.
 - o Delete the corresponding executable file from its path.
2. If unsure: o Right-click → Suspend Process to stop it temporarily for investigation.
3. After removal:
 - o Run a Full System Scan using Windows Defender or Malwarebytes to ensure no remnants remain.



Step 7: Example Observation

You find faangpath_simple_template.pdf consuming 70% CPU.

- Path: C:\Users\Admin\AppData\Temp\faangpath_simple_template.pdf
- Digital Signature: None
- Company Name: Unknown
- Network Activity: Shows connections to unknown IPs in the TCP/IP tab
- Online Check: VirusTotal confirms it as a known trojan
- Action Taken: Suspended → Killed → Deleted file → Performed full antivirus scan



Result:

Using Process Explorer, suspicious processes were successfully identified by examining their CPU usage, path, digital signature, and network activity.

Confirmed malicious processes were terminated and removed to maintain system integrity.
